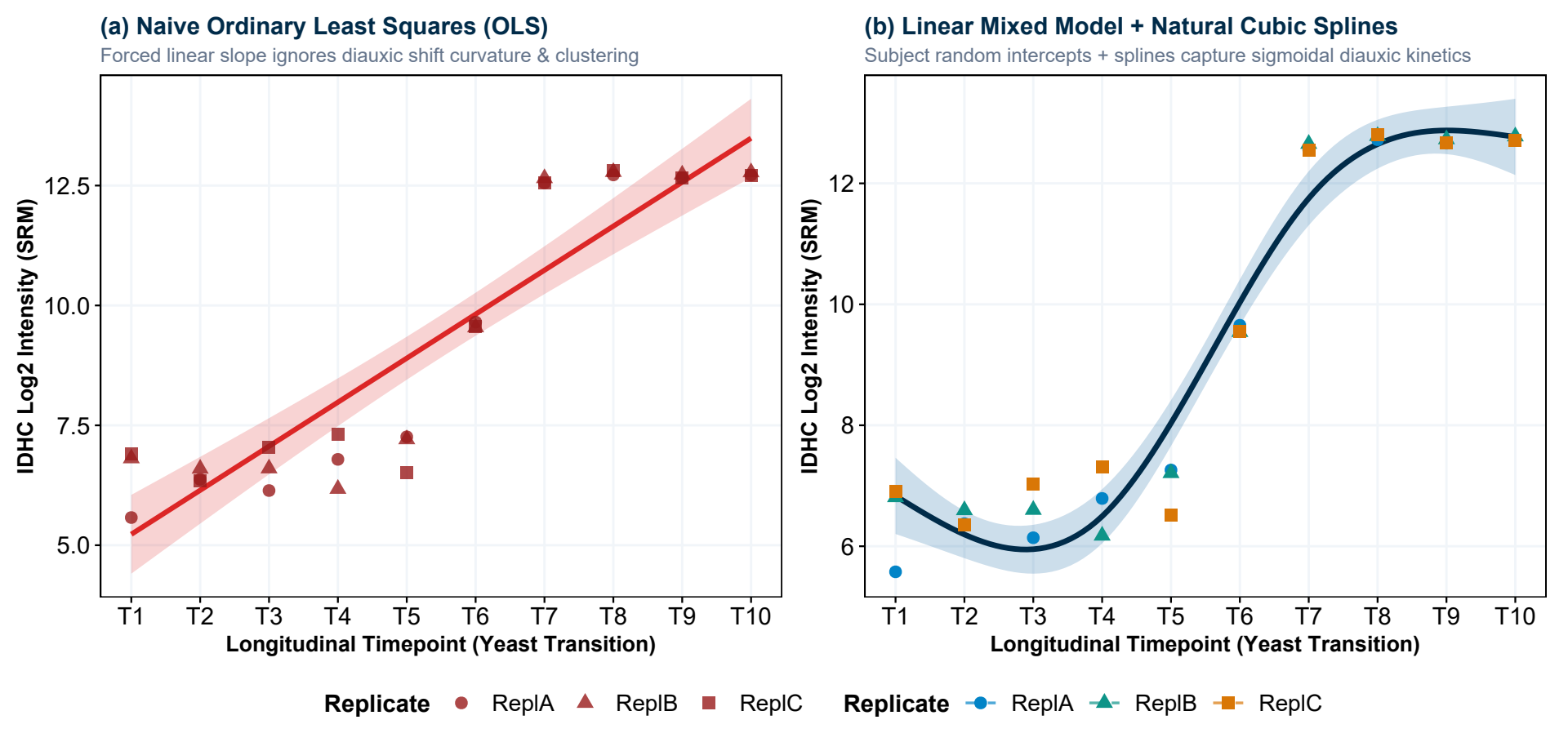
(a) Naive Ordinary Least Squares (OLS)
Forced linear slope ignores diauxic shift curvature & clustering
IDHC Log2 Intensity (SRM)
12.5
10.0
7.5
5.0
T1
T2
T3
T4
T5
T6
T7
T8
T9
T10
Longitudinal Timepoint (Yeast Transition)
(b) Linear Mixed Model + Natural Cubic Splines
Subject random intercepts + splines capture sigmoidal diauxic kinetics
IDHC Log2 Intensity (SRM)
12
10
8
6
T1
T2
T3
T4
T5
T6
T7
T8
T9
T10
Longitudinal Timepoint (Yeast Transition)
Replicate
ReplA
ReplB
ReplC
Replicate
ReplA
ReplB
ReplC